

Solution for Chapter 5

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5.1

(a) When $p = 2$,

$$\mathcal{L}(\mathbf{W}, b, \alpha, \beta, \xi) = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^m \xi_i^2 - \sum_{i=1}^m \alpha_i [y_i(\mathbf{w}\mathbf{x}_i + b) - 1 + \xi_i] - \sum_{i=1}^m \beta_i \xi_i$$

Setting the gradient of the Lagrangian with respect to the $\mathbf{w}, b, \xi_i$ is to zero, the results we got is same as (5.26), (5.27), (5.29), (5.30). (5.28) is replaced by $2C\xi_i = \alpha_i + \beta_i$.

$$\begin{aligned} \inf \mathcal{L} &= \frac{1}{2} \left\| \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i \right\|^2 + C \sum_{i=1}^m \left(\frac{\alpha_i + \beta_i}{2C} \right)^2 - \sum_{i=1}^m \alpha_i y_i b + \sum_{i=1}^m \alpha_i - \sum_{i=1}^m \alpha_i \xi_i - \sum_{i=1}^m \beta_i \xi_i - \sum_{i=1}^m \alpha_i y_i \mathbf{w}\mathbf{x}_i \\ &= \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \alpha_i \alpha_j y_i y_j \mathbf{x}_i \mathbf{x}_j - \frac{1}{4} \sum_{i=1}^m \frac{(\alpha_i + \beta_i)^2}{C} \end{aligned}$$

(b) Yes, this problem is still convex because the first two items are convex, and $-\frac{1}{4} \sum_{i=1}^m \frac{(\alpha_i + \beta_i)^2}{C}$ is concave but we can use convec techniques to solve it.

4.2